# Light Based Applications in Medicine: Integrated Pulse Oximetry-Glucose Monitoring

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***Abstract*—The convergence of pulse oximetry and glucose monitoring technologies represents a paradigm shift in non-invasive biosensing. This review examines the technical requirements, current research progress and implementation challenges of integrated optical systems capable of simultaneous oxygen saturation (SpO) and blood glucose level (BGL) measurements. We analyze recent advances in photoplethysmography (PPG) signal processing, multiwavelength optical configurations and machine learning algorithms applied to glucose estimation. Also highlighted is the necessity of multi-spectral light sources, photodetectors with broad spectral sensitivity and thorough signal processing pipelines for extracting weak glucose signatures from physiological noise. While technical and regulatory hurdles remain, integrated pulse oximetry-glucose monitors show incredible potential for diabetes monitoring and management.**

***Index Terms*—Photoplethysmography, Beer-Lambert Law, Oxygen saturation (SpO), Blood Glucose Level (BGL)**

## I. Introduction

The global diabetes epidemic affects approximately 589 million people, with projections reaching 853 million by 2050 [1]. Continuous Glucose Monitoring (CGM) is crucial for preventing complications associated with diabetes. Conventional methods remain invasive, requiring blood extraction via finger-pricking or subcutaneous implants. This causes discomfort, infection risk, and patient non-compliance. In this paper we propose a hybrid optical system that augments traditional SpO measurement (using the pulse oximeter) with non-invasive glucose estimation, using photoplethysmography. Pulse oximetry measures arterial oxygen saturation $(SpO)$ by analyzing the differential absorption of light through haemoglobin, providing an early indicator of hypoxia (SpO 90%), a significant morbidity if untreated. The Beer-Lambert law offers the theoretical foundation for photoplethysmography (PPG) and its application in the pulse oximeter. Observing that glucose exhibits distinct absorption rates in the Near Infrared (NIR) (1000nm – 2500nm) motivates the integration of glucose monitoring alongside oxygen saturation using Beer-Lambert Law [2]. However, glucose optical signatures are orders of magnitude weaker than haemoglobin and masked by confounding factors including tissue scattering, melanin content and physiological noise. We examine meta components required for such integrated systems, analyze current research prototypes and evaluate their clinical viability. Extended Beer–Lambert Law: We present a generalized form of the Beer–Lambert incorporating multiple chromophores.

$$A_\lambda = \sum(\epsilon_{\lambda,i} \cdot c_i \cdot L) + G \tag{1}$$

where $A_\lambda \rightarrow$ absorbance at wavelength $\lambda$
$\epsilon_{\lambda,i} \rightarrow$ extinction coeff. of analyte $i$ (Glucose, HbO, Hb)
$c_i \rightarrow$ concentration
$L \rightarrow$ optical path length
$G \rightarrow$ scattering losses

## II. Related Work

Non-invasive glucose monitoring has been a focus of research for decades, with photoplethysmography (PPG) emerging as a promising technique due to its compatibility with existing pulse oximetry hardware [3]. Early studies leveraged the Beer-Lambert law to estimate glucose concentrations via differential absorption in the NIR spectrum, but challenges such as low signal-to-noise ratios limited accuracy [4]. Recent advances have incorporated multiwavelength PPG to capture glucose-specific absorption peaks beyond the standard red (660 nm) and infrared (940 nm) used in SpO monitoring [5]. For instance, systems using wavelengths like 1020 nm, 1300 nm, 1600 nm, and 2100-2330 nm have shown improved sensitivity to glucose signatures [6]. A proof-of-concept in-ear PPG device demonstrated feasibility by integrating a single NIR LED (880 nm) with machine learning for continuous monitoring, achieving correlations with invasive measurements [7]. Machine learning has further enhanced estimation accuracy. Deep learning models, such as convolutional neural networks (CNNs) and long short-term memory (LSTM) hybrids, process raw PPG signals to predict BGL without explicit feature extraction, outperforming traditional methods in noisy environments [8]. One study reported root mean square error (RMSE) values as low as 19.7 mg/dL using segmented 1-second PPG windows [9]. Federated learning approaches have addressed data privacy in large-scale datasets, enabling non-invasive monitoring across diverse populations [10]. However, prototypes often face calibration issues and variability due to skin tone, motion artifacts, and environmental factors [11]. Comprehensive reviews highlight the need for broader spectral sensitivity and advanced signal processing to overcome these hurdles [12] [13].

## III. Proposed Technological Layout

The proposed system integrates SpO and BGL monitoring into a single wearable device, leveraging expanded optical and electronic subsystems. This layout builds on standard pulse oximetry while addressing glucose-specific challenges through multiwavelength sources, advanced photodetectors, and intelligent signal processing.

### A. Optical Subsystems

Multiwavelength sources extend beyond standard LEDs (600 nm red and 900 nm IR) to include glucose-targeting wavelengths such as 1020 nm, 1300 nm, 1600 nm, and 2100-2330 nm, where absorption peaks are prominent [5]. Laser diodes or vertical-cavity surface-emitting lasers (VCSELs) provide precise, tunable emission for deeper tissue penetration and reduced power consumption compared to broadband sources. The system employs reflectance mode configurations, enabling measurements at diverse sites (e.g., finger, earlobe, wrist) with improved penetration depth over transmission mode. Tinkering with optical paths, such as variable source-detector spacing, enhances signal quality by minimizing scattering losses.

### B. Broad Spectrum Photodetectors

Conventional silicon photodiodes (800-1100 nm range) are supplemented with indium gallium arsenide (InGaAs) detectors (900-1700 nm) or extended InGaAs (900-2500 nm) to capture glucose-specific signatures [6]. This broadens spectral sensitivity, allowing differentiation of weak glucose absorbance from hemoglobin and water interference.

### C. Electronic Subsystems

High-resolution signal acquisition uses 16-24 bit analog-to-digital converters (ADCs) to resolve subtle glucose-related signals from noise. Advanced filtering eliminates motion artifacts, ambient light interference, and physiological noise (e.g., respiration, vasomotion). Multi-channel control employs time-division or frequency-division multiplexing to coordinate multiple light sources, preventing spectral interference.

### D. Feature Extraction and Machine Learning

Feature extraction derives parameters from PPG waveforms, including pulse shapes, amplitudes, area ratios, frequency-domain metrics, and phase relationships between hemodynamic parameters [14]. Deep learning models, such as CNNs for end-to-end glucose prediction from raw PPG, or CNN-LSTM hybrids for spatial-temporal feature extraction, bypass explicit feature engineering [8]. Signal segmentation into 1-10 second PPG windows improves real-time performance while maintaining accuracy (e.g., RMSE: 19.7 mg/dL) [9]. Empirical mode decomposition (EMD) further refines signals by decomposing them into intrinsic mode functions for noise reduction [15].

## IV. Implementation Challenges and Clinical Viability

Key challenges include signal confounding from tissue variability, requiring adaptive calibration algorithms [11]. Regulatory hurdles, such as FDA approval for accuracy in diverse populations, demand rigorous clinical trials [2]. Globally, the clinical viability of integrated systems is supported by ongoing advancements, but local contexts must be considered. In Kenya, where diabetes prevalence stands at approximately 3.1% among adults, affecting over 813,300 individuals, the demand for affordable, non-invasive monitoring is particularly high due to limited healthcare resources and rising cases [16] [17]. Kenyan clinical viability is enhanced by feasibility studies demonstrating CGM use in rural and urban settings, with trials showing improved metabolic control and patient acceptance [18] [19] [20]. However, challenges include variability in skin tones and environmental factors prevalent in diverse Kenyan populations. Regarding Kenyan regulations, medical devices, including non-invasive glucose monitors, must be registered with the Pharmacy and Poisons Board (PPB) through their online portal, following guidelines for evaluation and registration [21] [22] [23]. The PPB classifies devices and requires compliance with international standards, with pathways for fast-track approval for innovative technologies. Clinical trials in Kenya require approval from the PPB and local ethics committees, ensuring safety and efficacy in the local context [24]. Ongoing projects like the ACCEDE initiative aim to improve access to CGM devices, highlighting regulatory support for such innovations [25] [26]. Prototypes like wristband or in-ear devices show promise, but scalability requires low-cost components and edge computing for real-time processing [7] [27]. Future work could integrate microwave sensing for complementary data [12].

## V. Summary

The proposed integrated pulse oximetry-glucose monitoring system offers a non-invasive, convenient solution for diabetes management. By leveraging multiwavelength PPG and AI, it addresses current limitations, paving the way for widespread adoption. Ongoing research must focus on validation to realize its full potential, particularly in regions like Kenya where tailored regulatory and clinical approaches can accelerate implementation.

## VI. Prototype Implementation – Proposition for how to build

### A. Complete Bill of Materials (Nairobi considerate prices)

### B. Hardware Architecture (Build Instructions)

- Optical layout: Reflectance mode finger clip – eight LEDs arranged in a ring 8 mm from the central InGaAs photodiode.
- LED driving: N-channel MOSFETs + ESP32 GPIO for time-division multiplexing (50 µs pulses, 500 Hz per wavelength).

TABLE I

Full BOM – Single prototype (prices from Nerokas, Luthuli Avenue, AliExpress with duty)

| Item | Source | Price (KSh) |
|---|---|---|
| ESP32-S3-DevKitC-1-N8R8 (8 MB Flash, 8 MB PSRAM) | Nerokas / Jumia | 1 800 |
| MAX30102 (660 nm + 880 nm) – standard $SpO_2$ | Nerokas | 450 |
| APD1300-04 InGaAs photodiode (900–2100 nm, 4 $mm^2$) | AliExpress (Hamamatsu clone) | 1 200 |
| SFH 4545 1050 nm / 1070 nm IR LED (×2) | AliExpress | 400 |
| LED1300-10-120 1300 nm LED (×2) | AliExpress / LaserComponents | 650 |
| LED1450-03 1450 nm LED (×1) | AliExpress | 450 |
| LED1600-03 1600 nm LED (×1) | AliExpress | 550 |
| LED1950-03 1950 nm LED (×1) | AliExpress | 750 |
| OPA2388 zero-drift transimpedance amplifier | AliExpress | 350 |
| 0.96” OLED SSD1306 I2C | Nerokas | 450 |
| TP4056 + 500 mAh LiPo + switch | Jumia | 350 |
| 3D-printed finger clip (black PLA/TPU) | Local fablab | 600 |
| Custom PCB (10×15 cm, JLCPCB 5 pcs) | JLCPCB + shipping | 1 200 |
| Misc (resistors, MOSFET drivers, connectors) | Luthuli | 400 |
| **Total one-off prototype** | | **KSh 7 550** |
| **Total Estimate at 100-unit volume** | | **KSh 4 200** |

- Amplification: OPA2388 configured as transimpedance amp (gain 1 M–100 M switchable via digital potentiometer).
- ADC: ESP32-S3 internal 12-bit SAR ADC (sufficient after amplification) or optional ADS1115 for 16-bit.
- Power: 3.3 V rail, total consumption 80 mA during measurement, 48 h on 500 mAh battery.

### *C. Firmware and On-Device AI*

- Framework: ESP-IDF v5.2 with TensorFlow Lite Micro / Edge Impulse.
- Signal chain: 500 Sa/s per channel → DC removal → bandpass 0.5–10 Hz → beat detection → per-wavelength AC/DC extraction.
- Glucose model: 3.1M-parameter 1D ResNet (input: 8×500 PPG window) train on 500 Kenyan volunteers (mix skin tones, OGTT synchronised). Sensible validation aim: Clarke Error Grid 92% Zone A+B, MARD 21.4 mg/dL.
- $SpO_2$ calculation: Standard ratio-of-ratios using 660/940 nm.
- Output: OLED shows $SpO_2$, heart rate, and estimated glucose every 60 seconds.

### *D. 3D-Printed Mechanical Design*

- Finger clip: Spring-loaded, black PLA body + soft TPU pad.
- Optical shielding: Internal baffles + matte black paint to reduce ambient light and cross-talk.
- STL files included in repository.

### *E. Local Sourcing and Assembly Notes for Kenyan Builders*

- 660/880 nm: Use any MAX30102 module – desolder if needed.
- 1050–1950 nm LEDs: Search AliExpress for exact part numbers above; typical delivery 2–3 weeks.
- InGaAs photodiode: Cheapest reliable source is ”G12180-010A” Hamamatsu equivalent – KSh 1 200 delivered.
- PCB: Order from JLCPCB (5-day express), solder at any Luthuli Avenue electronics bench (KSh 500 labour) you can also check Kenyatta University Chandaria Building (access to utilities completely free) to use their solder bench.
- Calibration: Requires one-time venous blood reference (KSh 300 at any Level-4 clinic).

## VII. Epilogue

Every part is buyable in Nairobi today. With KSh 8 000 and a few weeks, any final-year EEE or BME student at Kenyatta University, JKUAT, Strathmore, or Technical University of Kenya can build a device that simultaneously measures oxygen saturation and estimates blood glucose non-invasively. Unangoja nini?

We invite innovation, independent clinical validation at Kenyan hospitals, and contributions to build this new exciting technology. Together we can break the KSh 30 000/month CGM barrier and bring continuous glucose awareness to every Kenyan with diabetes.

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